# Seyed Ali Padash Sotoodeh

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## EDUCATION

**BSc in Computer Engineering** Sep 2021 – Feb 2026
*K. N. Toosi University of Technology, Tehran, Iran*
Final grade: **18.37 / 20**
Thesis: *Sentiment Analysis of User Comments Using Artificial Intelligence* (grade: 19.5 / 20)

**Secondary School Diploma – Mathematics and Physics** Sep 2018 – Jun 2021
*Professor Samii High School, Rasht, Iran*
Final grade: **19.38 / 20**

## WORK EXPERIENCE

**Head Teaching Assistant** Jul 2025 – Aug 2026
*Sejong University, Seoul, South Korea (remote) · Supervisor: Prof. Sadeghi-Niaraki*

- Led a team of 6 TAs producing lecture slides, course materials and instructional videos for *Basic Machine Learning* and *Mathematics for Deep Learning*; designed the workflow and reviewed the team's output.
- Built a Python tool that turns Jupyter notebooks into Colab-style animated lectures (code typed on cue, outputs revealed step by step, LaTeX rendering), replacing live typing during recording.

**Teaching Assistant – Fundamentals of Programming** Winter semesters 2022/23 and 2024/25
*K. N. Toosi University of Technology*

- Designed C coding exercises, revised lesson documents and mentored a student group.

**Python Developer (credited university internship)** Jul 2024 – Aug 2024
*K. N. Toosi University of Technology*

- Refactored a Python web crawler into modular, SOLID-based classes with logging, and extended it with multiprocessing, SQL process tracking and a Django + React interface.

## PROJECTS

**ChamFlick DE: German Learning App for Books and Films (Android)** Aug 2026 – present
ali-padash.github.io/chamflick-showcase

- Offline Android app for reading German: imports books, scanned pages and film subtitles, colours every noun by gender and number, and opens a dictionary card when a word is tapped, splitting compounds into their parts.
- Runs a German sentence parser on the phone (spaCy's model, re-implemented in C++), so split and reflexive verbs are looked up under the right base form.

**YouTube Comment Sentiment Analysis (BSc Thesis)** Jun 2025 – Oct 2025

github.com/ali-padash/youtube-comment-sentiment-analysis

- Improved lightweight classifiers (Logistic Regression, Linear SVM) for 3-class YouTube comment sentiment: raised their macro F1 on the original labelled test data from 66.4% / 65.1% to about 72.2%.
- Combined VADER lexicon features with weakly supervised data: YouTube comments collected via the YouTube Data API and pseudo-labelled with Twitter-RoBERTa, which outperformed paraphrase augmentation with Gemma 3.
- Built a Streamlit app that labels comments from the API or a CSV, with a hybrid mode that runs the fast model and falls back to Twitter-RoBERTa only when its confidence is low.

**Rahi: Bus Ticket Reservation System (Full-Stack Web App)** Mar 2024 – May 2024; extended Oct 2026

github.com/ali-padash/bus-ticket-reservation-system

- Built a bus booking site modelled on an Iranian booking service, with trip search and a seat map: React + TypeScript frontend, Go backend (Gin, GORM) on PostgreSQL.
- Extended in Oct 2026 with ticket booking (10-minute seat hold), a bilingual English/Persian mobile-friendly interface, CI and Docker.
- Prevented double-booking with row-level locks and a partial unique index, verified by concurrent tests (20 simultaneous buyers for one seat, exactly one succeeds).

### Further projects

**8-bit CPU** · Verilog, ModelSim

**Reliable transfer over UDP** · Go-Back-N, CRC

**Genetic algorithm in hardware** · SystemC

**Persian news classification** · NLP, ranked 2nd of 18

Details and more at ali-padash.github.io

## TECHNICAL SKILLS

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|---|---|
| **Programming:** | Python, C++, C, Java, Kotlin, TypeScript/JavaScript, Go, SQL (PostgreSQL) |
| **ML & NLP:** | PyTorch, scikit-learn, Transformers, spaCy, pandas, NumPy, OpenCV, Streamlit |
| **Web:** | Django, React, Node.js/Express, Gin |
| **Hardware Design:** | Verilog, VHDL, SystemC, ModelSim |
| **Tools:** | Git, GitHub Actions, Docker, LaTeX |

## LANGUAGES

**Persian:** Native · **English:** C1 · **German:** B1

## INTERESTS

Music (composing in FL Studio) · Learning German through films and comics